

1. INTRODUCTION

The stock market or equity market comprises several stock ex-changes worldwide. Investors and the general public buy and sell shares that constantly switch their prices due to the law of demand and supply. A share or stock is ownership in the firm or corporation. Buyers try to buy a share at the lowest possible price, while sellers try to sell it at the highest possible price. The largest stock exchange in the world is the New York Stock Exchange (NYSE), with as of February 2018; it had market capitalization worth US \$30.1 trillion. The stock market is one of the most important platforms to raise money, along with debt markets which are more imposing but do not trade publicly. The stock exchange is highly liquid, making it easy for the interested parties to sell or buy securities easily. A key feature in any upcoming economy is the increased involvement of the people in the stock market and its upward movement, too.

The financial market is a dynamic and composite system where people can buy and sell currencies, stocks, equities and derivatives over virtual platforms supported by brokers. The stock market allows investors to own shares of public companies through trading either by exchange or over the counter markets. This market has given investors the chance of gaining money and having a prosperous life through investing small initial amounts of money, low risk compared to the risk of opening new business or the need of high salary career.

Stock markets are affected by many factors causing the uncertainty and high volatility in the market. Although humans can take orders and submit them to the market, automated trading systems (ATS) that are operated by the implementation of computer programs can perform better and with higher momentum in submitting orders than any human. However, to evaluate and control the performance of ATSs, the implementation of risk strategies and safety measures applied based on human judgements are required. Many factors are incorporated and considered when developing an ATS, for instance, trading strategy to be adopted, complex mathematical functions that reflect the state of a specific stock, machine learning algorithms that enable the prediction of the future stock value, and specific news related to the stock being analysed.

ANN or Artificial Neural Network is an algorithm designed to understand complicated issues that cannot be undertaken by basic machine learning algorithms or

accessible neural networks. ANNs are built in a more complex and complicated web of interconnectivity than the human brain. The method is based on algebraic equations aiming to channel information to a model or time-series line. SVM or Support Vector Machines is a supervised machine learning model which can either be used for classification or regression challenges. Being most extensively used for classification problems, the SVM algorithm works best in spaces of high dimension or in situations where several dimensions exceed the number of samples. This algorithm provides abnormal returns on investments to investors and serves as a risk management tool. The last machine learning algorithm discussed in this paper is LSTM or Long Short-Term Memory Networks. LSTM, a popular recurrent neural network, can learn order dependence for solving sequence prediction problem. The LSTM algorithm comprises the cell, input, output, and forgets gate. The gates look after the inflow of data in and out of the cell. The cell is the primary unit of the model, which remembers values over various time intervals. Time-series prediction is a common technique widely used in many real-world applications such as weather forecasting and financial market prediction. It uses the continuous data in a period of time to predict the result in the next time unit. Many time series prediction algorithms have shown their effectiveness in practice. Stock market is a typical area that presents time-series data and many researchers' studies on it and proposed various models. In this project, Time series models are used to predict the stock price.

Existing System:

Most research with machine learning forecasting has focused on Artificial Neural Networks (ANN). ANNs have a series of interconnected nodes that simulate individual neurons, and are organized into different layers based on function (input layer, processing layer, output layer, etc.). The ANN assigns weights to connections, and the output is calculated based on the inputs and the weights. Although ANNs are accurate, when the data does not have sudden variations. And as the stock market data varies so greatly with time and non-linearly it has been proven that prediction is difficult even with advanced techniques like ANNs.

Proposed System:

In the proposed system, we use support vector machines (SVM) to deal with the problem of accurate prediction due to volatility, which can be seen in artificial neural networks (ANNs). The new technique, support vector machines, aims to adapt to volatility and produce much more accurate results.

Purpose of the project:

- The Main Objective of this Project is to find out the Predictive Capabilities of SVM model on stocks.
- And see if SVM Model is Practical for every day stock prediction by taking a recent dataset.

2. LITERATURE SURVEY

The applications of artificial neural networks, support vector machines, and long–short term memory for stock market prediction:

In the simplest of words, Artificial Neural Networks are a mesh of numerical equations. One or more input variables are taken and then processed by a sequence of equations resulting in one or multiple outputs. Any network generally has three layers — an input layer, Hidden layer, and Output layer. The layer containing all the feature variables shown as x_1 , x_2 , and x_3 up to x_n is the input layer. The hidden layer comprises one or more nodes (hidden units). The circles in the diagrams below represent a node. Then the output, which can be one or more than one. The more nodes and layers, the more complex calculations can be solved through the network. An artificial neural network structure containing four neurons in the first layer and a single neuron in the second layer is an example of a generic structure. The hidden layer is considered the most crucial layer of the algorithm; it acts as a distillation layer that only forwards important data and patterns from the input to the next layer. It makes the network significantly faster and efficient. Represents the neural network layers of three types of layers: input layer, hidden layer and output layer. The key to getting a good model is the accurate prediction of the Weights. The Back Propagation Algorithm achieves this task; this algorithm is what makes ANN a learning model. It learns from the mistakes and adjusts itself accordingly. ANN can model better data with high volatility and non-constant variance. ANN has also come out as the most efficient in predicting financial time series as the Data is also highly volatile. Shahvaroughi Farahani and Razavi Hajiagha. Today, the stock market serves an essential purpose and may be used to indicate economic strength. To forecast stock price index values using an artificial neural network (ANN) that has been trained utilizing many novel meta-heuristic algorithms like social spider optimization (SSO) as well as the bat algorithm (B.A.) is described. As input variables, some technical indicators were used. They then utilized genetic algorithms (G.A.) as a heuristic method for feature selection and indication selection. But in the other hand, specific time series forecasting models, such as ARMA and ARIMA, are also used to forecast stock prices. It finds that via the use of G.A., the set of input variables was substantially decreased. [1]

As a result, the computation speed and accuracy of the network and the coefficient of determination improved. Also, the Hybrid models achieve a higher level of accurately explaining the model. As a result, the primary suggestion is to train the network using various novel meta-heuristic methods. Jamous et al. As mentioned before, artificial neural networks (ANNs) have already been used to forecast stock market closing prices. However, standalone ANNs have several constraints, which result in reduced prediction accuracy. This constraint is overcome via the use of hybrid models. As a result, the literature reported on a mixture of artificial intelligence networks & particle swarm optimization for effective stock market prediction. To accomplish high-accuracy prediction in a short time, this article proposes a new enhanced technique dubbed PSOCOG. Hence ANN-PSOCOG beat ANN-SPSO in prediction accuracy by about 13%, SPSOCOG by about 17%, SPSO by nearly 20%, and ANN by nearly 25% using the S&P 500 dataset. When the DJIA dataset was used, ANN-PSOCOG beat ANN-SPSO by about 18%, SPSOCOG by approximately 24 percent and SPSO by approximately 33 percent and ANN by approximately 42 percent in terms of prediction accuracy. Additionally, the suggested model is tested in the presence of COVID-19. The findings demonstrated the suggested model's ability to accurately forecast the closing price when MAPE, MAE, and RE values were extremely tiny for the S&P 500, GOLD, NASDAQ-100, & CANUSD datasets. Selvamuthu et al. conducted a study to remove the barrier of time series data by using the ANN algorithm. The algorithm incorporates analysing and hence predicting stock exchanges' price movements. The aim was to find if the algorithm would forecast stock prices despite its dynamic nature and liable to quick changes. The abundance of data and technological enhancement makes it possible to form algorithms that can work and how one wish. The study employs three algorithms: Levenberg–Marquardt (L.M.), Scaled Conjugate Gradient (SCG), and Bayesian Regularization. The results showed that SCG gave the best validation in 103 (54) iterations and L.M. in 10 tick datasets. However, referring to the time barrier again, SCG was quicker than L.M. Bayesian Regularization gave the least squared error over all datasets. Nevertheless, performance-wise Scaled Conjugate Gradient was the most promising. The study used three Neural Network learning algorithms. The three algorithms provided 99.9% accuracy when using tick data. Moghaddam et al. experimented with the ability of ANN to predict the NASDAQ. Two networks for the NASDAQ index prediction were developed and validated. The methods

used in the study took into context both short-term and historical stock prices along with daily data. ANN has performed better in bankruptcy prediction, discriminate analysis, and logistic regression. ANN can predict better than statistical methods as it is related complexly among input variable data and financial data. The model used input parameters as previous four to nine days (working). The model output displayed nothing to do with the number of days as inputs in the prediction process. Trifonov et al. examined the ANN algorithm under the contract with National Science Fund in Bulgaria. The research included extensive experimenting with the network parameters. They could confidently state that choice of input, pre-processing of data, variables, and the network's structure had a significant impact on the working of the algorithm. After running the test multiple times, the researchers concluded that the Neural Network Algorithm is considerably accurate in predicting the movement of the prices a day before the actual price change. Systems designed with Neural Networks are universal and can be applied to any number of financial devices using the most basic technical indicators as input data. There are three conventional approaches for stock price prediction: technical analysis, traditional time series forecasting, and machine learning method. Earlier classical regression methods such as linear regression, polynomial regression, etc. were used to predict stock trends. Also, traditional statistical models which include exponential smoothing, moving average, and ARIMA make their prediction linearly. Nowadays, Support Vector Machines (Cortes & Vapnik, 1995) (SVM) and Artificial Neural Networks (ANN) are widely used for the prediction of stock price movements. Every algorithm has its way of learning patterns and then predicting. Artificial Neural Network (ANN) is a popular and more recent method which also incorporates technical analysis for making predictions in financial markets. ANN includes a set of threshold functions. These functions trained on historical data after connecting each other with adaptive weights and they are used to make future predictions. (Trippi & Turban, 1992; Walczak, 2001; Shadbolt & Taylor, 2002) (Kuan & Liu, 1995) investigated the out-of-sample forecasting ability of recurrent and feed forward neural networks based on empirical foreign exchange rate data (Kuan & Liu, 1995). In 2017, Mehdi Khashei and Zahra Haji Rahimi evaluated the performance of series and parallel strategies to determine a more accurate one using ARIMA and MLP (Multilayer Perceptron) (Mehdi & Zahra, 2017).

Indian Stock market prediction using artificial neural networks on tick data:

Artificial neural networks have been used widely to solve many problems due to its versatile nature. (Samek & Varachha, 2013) (Yodele et al., 2012), presented a hybridized approach, i.e., a combination of the variables of fundamental and technical analysis of stock market indicators to predict future stock prices to improve the existing methods, (Yodele et al., 2012) (Y Kara & A Boyacioglu, 2011) discussed stock price index movement using two models based on Artificial Neural Network (ANN) and Support Vector Machine (SVM). They compared the performances of both the models and concluded that the average performance of the ANN model was significantly better than the SVM model. (Y Kara & A Boyacioglu, 2011) (Qi & Zhang, 2008) investigated the best modelling of trend time series using Neural Network. They used four different approaches, i.e., raw data, raw data with a time index, de-trending and differencing for modelling various trend patterns and concluded Neural Network gives better results (Qi & Zhang, 2008). H.K. Cigizoglu, (2003) discussed the application of ANN forecasting, estimation and extrapolation of the daily flow data belonging to the rivers in the East Mediterranean region of Turkey.

In their study, they found that ANN provides a better fit to the data than conventional methods (Cigizoglu, 2003). ANN can consider as a computation or a mathematical model which is inspired by the functional or structural characteristics of biological neural networks. These neural networks are developed in such a way that it can extract patterns from noisy data. ANN first train a system using a large sample of data known as training phase then it introduces the network to the data which was not included in the training phase, this phase known as validation or prediction phase. The sole motive of this procedure is to predict new outcomes. (Bishop, 1995) This idea of learning from training and then predicting outcomes in ANN comes from the human brain which can learn and respond. Thus ANN has been used in many applications and is proven successful in executing complex functions in a variety of fields (Fausett, 1994) [2].

G. F. Bjoern Krollner, Bruce Vanstone, “Financial time series forecasting with machine learning techniques: A survey,” in European Symposium on Artificial Neural Networks: Computational and Machine Learning, Bruges, Belgium, April 2010:

We put forward the deep neural network ensemble to model and predict Chinese stock market index (including Shanghai composite index and SZSE component index). The multi-layer feed forward deep neural networks trained by historical data performs as a supervised regressor to predict the daily high, low and close, based on the input of past several days' indices. The conventional sigmoid or tanh activation function is replaced with the recently proposed Leaky Rectified Linear Unit (LReLU) in neural networks, which enables deep network to be trained efficiently. The backpropagation algorithm and Adam optimization algorithm, which accelerates the training speed enormously, are applied in our neural networks [3].

A finite set of these deep neural networks (each neural network in the set is called a component neural network) are trained to form a neural network ensemble. Each component neural network is trained with a component training set bootstrapping from the original training set, and the average of the predictions of all component neural networks is the final output. That's called Bagging approach, a paradigm of ensemble learning, considered to reduce the generalization error significantly.

A. Kar, "Stock prediction using artificial neural networks," Dept. of Computer Science and Engineering, IIT Kanpur, 2016:

The historical data of Shanghai composite index and SZSE component index of the past 26 years is collected and pre-processed. Each dataset is divided into training period and test period. The training period is used to train the neural network ensembles and the test period is used to test the performance of the predictions. On test period, the relative error of predictions is computed, as well as the accuracy of trend predictions of index change, which are used to measure the performance of predictions.

Most research with machine learning forecasting has focused on Artificial Neural Networks (ANN) [4]. ANNs have a series of interconnected nodes that simulate individual

neurons, and are organized into different layers based on function (input layer, processing layer, output layer, etc.). The ANN assigns weights to connections, and the output is calculated based on the inputs and the weights. As the machine trains, it notices patterns in the training data and reassigns the weights. Kar demonstrates that ANNs are quite accurate when the data does not have sudden variations [5]. Patel and Yalamalle agree that ANNs can predict with accuracy slightly greater than 50%, but caution that since stock market data varies so greatly with time and nonlinearly, prediction is difficult even with advanced techniques like ANNs [6].

S. R. Y. Mayank Kumar B Patel, “Stock prediction using artificial neural network,” International Journal of Innovative Research in Science, Engineering, and Technology, 2014:

Recent research in the field has used another technique known as Support Vector Machines in addition to or as an alternative to ANNs. Whereas ANNs are models that try to minimize classification error within the training data, SVMs may make classification errors within training data in order to minimize overall error across test data. A major advantage of SVMs is that it finds a global optimum, whereas neural networks may only find a local optimum. See 2.4 for the mathematics behind SVMs.

Using the SVM model for prediction, Kim was able to predict test data outputs with up to 57% Accuracy, significantly above the 50% threshold [7]. Shah conducted a survey study on stock prediction using various machine learning models, and found that the best results were achieved with SVM [8]. His prediction rate of 60% agrees with Kim’s conclusion. Since most recent research has incorporated SVMs, this is the technique we use in our analysis.

3. ANALYSIS

3.1 System Requirements

This Chapter describes about the requirements. It specifies the hardware and software requirements that are required in order to run the application properly. The Software Requirement Specification (SRS) is explained in detail.

1. Hardware Requirements:

- Processor: i3 or better
- RAM: 4GB (Minimum) | 8GB (Recommended)
- Graphics: 2GB or better

2. Software Requirements

- Jupiter Notebook
- Any Search Engine/ Browser (Ex: Chrome)
- Yahoo Reliance (For Dataset)

3.2 Problem Statement

The idea to use Computer science techniques to predict stock market is not a new occurrence, many prediction models have focused on linear statistical time series models such as ARIMA, AutoRegressive Integrated Moving Average, and ARCH, AutoRegressive Conditionally Heteroscedastic, which are linear and nonlinear model respectively but with only suboptimal result to be shown. The Main Idea of this project is to see how the use of SVM, which is known for its linear and nonlinear problem-solving capabilities [9], can better predict the stocks.

3.3 Stock Prediction

While making stock predictions, it is important to have a dataset of adequate quality for training and testing the machine learning model. And with the rise in artificial technology, people are turning their heads towards ways in which machine learning, which is a subset of artificial intelligence, can be used for building prediction models. The main idea is to have a predictive variable that is used for forecasting. This predictive variable is a Boolean value that signals if a stock should be bought or not. The way this predictive variable works is by checking the opening price and closing price on a particular day. If the open price is higher than the closing price, then the predictive variable should be set to 0 or 1. It is this very predicting variable that is used to train the machine learning model and test the model. After Training the model the model's accuracy is found to evaluate its performance.

4. Methodology

4.1 Supervised Learning Technique

Supervised learning uses a training set to teach models to yield the desired output. This training dataset includes inputs and correct outputs, which allow the model to learn over time. The algorithm measures its accuracy through the loss function, adjusting until the error has been sufficiently minimized. [10]

Supervised learning can be separated into two types of problems when data mining—classification and regression:

- **Classification** uses an algorithm to accurately assign test data into specific categories. It recognizes specific entities within the dataset and attempts to draw some conclusions on how those entities should be labelled or defined. Common classification algorithms are linear classifiers, support vector machines (SVM), decision trees, k-nearest neighbour, and random forest, which are described in more detail below.
- **Regression** is used to understand the relationship between dependent and independent variables. It is commonly used to make projections, such as for sales revenue for a given business. Linear regression, logistical regression, and polynomial regression are popular regression algorithms.

4.1.1 Supervised learning algorithms

Various algorithms and computations techniques are used in supervised machine learning processes. Below are brief explanations of some of the most commonly used learning methods, typically calculated through use of programs like R or Python:

- **Neural networks:** Primarily leveraged for deep learning algorithms, neural networks process training data by mimicking the interconnectivity of the

human brain through layers of nodes. Each node is made up of inputs, weights, a bias (or threshold), and an output. If that output value exceeds a given threshold, it's "fires" or activates the node, passing data to the next layer in the network. Neural networks learn this mapping function through supervised learning, adjusting based on the loss function through the process of gradient descent. When the cost function is at or near zero, we can be confident in the model's accuracy to yield the correct answer.

- **Naive bayes:** Naive Bayes is classification approach that adopts the principle of class conditional independence from the Bayes Theorem. This means that the presence of one feature does not impact the presence of another in the probability of a given outcome, and each predictor has an equal effect on that result. There are three types of Naïve Bayes classifiers: Multinomial Naïve Bayes, Bernoulli Naïve Bayes, and Gaussian Naïve Bayes. This technique is primarily used in text classification, spam identification, and recommendation systems.
- **Linear regression:** Linear regression is used to identify the relationship between a dependent variable and one or more independent variables and is typically leveraged to make predictions about future outcomes. When there is only one independent variable and one dependent variable, it is known as simple linear regression. As the number of independent variables increases, it is referred to as multiple linear regression. For each type of linear regression, it seeks to plot a line of best fit, which is calculated through the method of least squares. However, unlike other regression models, this line is straight when plotted on a graph.
- **Logistic regression:** While linear regression is leveraged when dependent variables are continuous, logistic regression is selected when the dependent variable is categorical, meaning they have binary outputs, such as "true" and "false" or "yes" and "no." While both regression models seek to understand relationships between data inputs, logistic regression is mainly used to solve binary classification problems, such as spam identification.
- **Support vector machines (SVM):** A support vector machine is a popular supervised learning model developed by Vladimir Vapnik, used for both

data classification and regression. That said, it is typically leveraged for classification problems, constructing a hyperplane where the distance between two classes of data points is at its maximum. This hyperplane is known as the decision boundary, separating the classes of data points (e.g., oranges vs. apples) on either side of the plane.

- **K-nearest neighbour:** K-nearest neighbour, also known as the KNN algorithm, is a non-parametric algorithm that classifies data points based on their proximity and association to other available data. This algorithm assumes that similar data points can be found near each other. As a result, it seeks to calculate the distance between data points, usually through Euclidean distance, and then it assigns a category based on the most frequent category or average. Its ease of use and low calculation time make it a preferred algorithm by data scientists, but as the test dataset grows, the processing time lengthens, making it less appealing for classification tasks. KNN is typically used for recommendation engines and image recognition.
- **Random forest:** Random Forest is another flexible supervised machine learning algorithm used for both classification and regression purposes. The "forest" references a collection of uncorrelated decision trees, which are then merged together to reduce variance and create more accurate data predictions.

4.1.2 Supervised learning examples

Supervised learning models can be used to build and advance a number of business applications, including the following:

- **Image- and object-recognition:** Supervised learning algorithms can be used to locate, isolate, and categorize objects out of videos or images, making them useful when applied to various computer vision techniques and imagery analysis.
- **Predictive analytics:** A widespread use case for supervised learning models is in creating predictive analytics systems to provide deep insights into various business data points. This allows enterprises to anticipate certain

results based on a given output variable, helping business leaders justify decisions or pivot for the benefit of the organization.

- Customer sentiment analysis: Using supervised machine learning algorithms, organizations can extract and classify important pieces of information from large volumes of data—including context, emotion, and intent—with very little human intervention. This can be incredibly useful when gaining a better understanding of customer interactions and can be used to improve brand engagement efforts.
- Spam detection: Spam detection is another example of a supervised learning model. Using supervised classification algorithms, organizations can train databases to recognize patterns or anomalies in new data to organize spam and non-spam-related correspondences effectively.

4.2 SVM

SVM is a powerful supervised algorithm that works best on smaller datasets but on complex ones. Support Vector Machine, abbreviated as SVM can be used for both regression and classification tasks, but generally, they work best in classification problems. They were very famous around the time they were created, during the 1990s, and keep on being the go-to method for a high-performing algorithm [10].

It is a supervised machine learning problem where we try to find a hyperplane that best separates the two classes. Don't get confused between SVM and logistic regression. Both the algorithms try to find the best hyperplane, but the main difference is logistic regression is a probabilistic approach whereas support vector machine is based on statistical approaches.

Now the question is which hyperplane does it select? There can be an infinite number of hyperplanes passing through a point and classifying the two classes perfectly. So, which one is the best? Well, SVM does this by finding the maximum margin between the hyperplanes that means maximum distances between the two classes.

4.2.1 Types of Support Vector Machine

- **Linear SVM**

When the data is perfectly linearly separable only then we can use Linear SVM. Perfectly linearly separable means that the data points can be classified into 2 classes by using a single straight line (if 2D).

- **Non-Linear SVM**

When the data is not linearly separable then we can use Non-Linear SVM, which means when the data points cannot be separated into 2 classes by using a straight line (if 2D) then we use some advanced techniques like kernel tricks to classify them. In most real-world applications we do not find linearly separable datapoints hence we use kernel trick to solve them.

Now let's define two main terms when dealing with support vector:

- **Support Vectors:** These are the points that are closest to the hyperplane. A separating line will be defined with the help of these data points.
- **Margin:** it is the distance between the hyperplane and the observations closest to the hyperplane (support vectors). In SVM large margin is considered a good margin. There are two types of margins **hard margin** and **soft margin**.

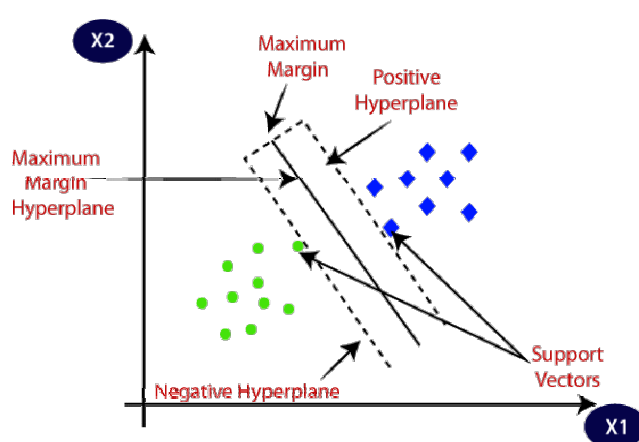


Fig 4.2.1.1 Support Vector Machine

4.2.2 Support vector workings

SVM is defined such that it is defined in terms of the support vectors only, we don't have to worry about other observations since the margin is made using the points which are closest to the hyperplane (support vectors), whereas in logistic regression the classifier is defined over all the points. Hence SVM enjoys some natural speed-ups.

Let's understand the working of SVM using an example. Suppose we have a dataset that has two classes (green and blue). We want to classify that the new data point as either blue or green.

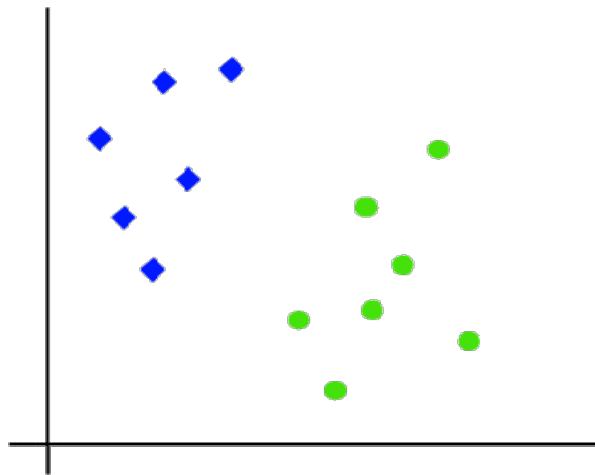


Fig 4.2.2.1 Clusters diagram

To classify these points, we can have many decision boundaries, but the question is which is the best and how do we find it? Since we are plotting the data points in a 2-dimensional graph we call this decision boundary a straight line but if we have more dimensions, we call this decision boundary a “hyperplane”

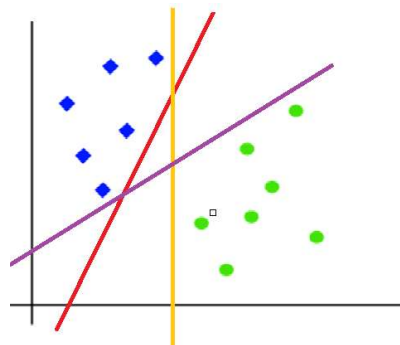


Fig 4.2.2.2 Different Hyperplanes

The best hyperplane is that plane that has the maximum distance from both the classes, and this is the main aim of SVM. This is done by finding different hyperplanes which classify the labels in the best way then it will choose the one which is farthest from the data points or the one which has a maximum margin.

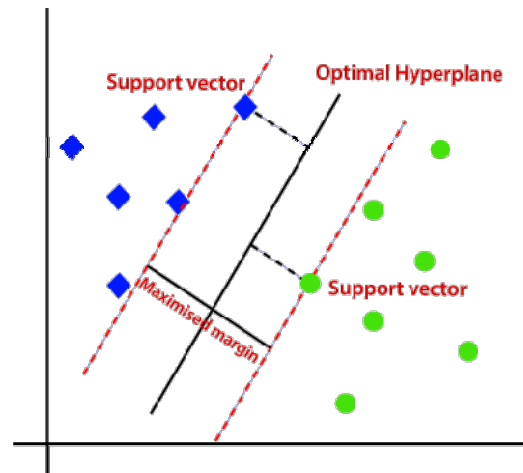


Fig 4.2.2.3 Best Hyperplane

5. ARCHITECTURE AND DESIGN

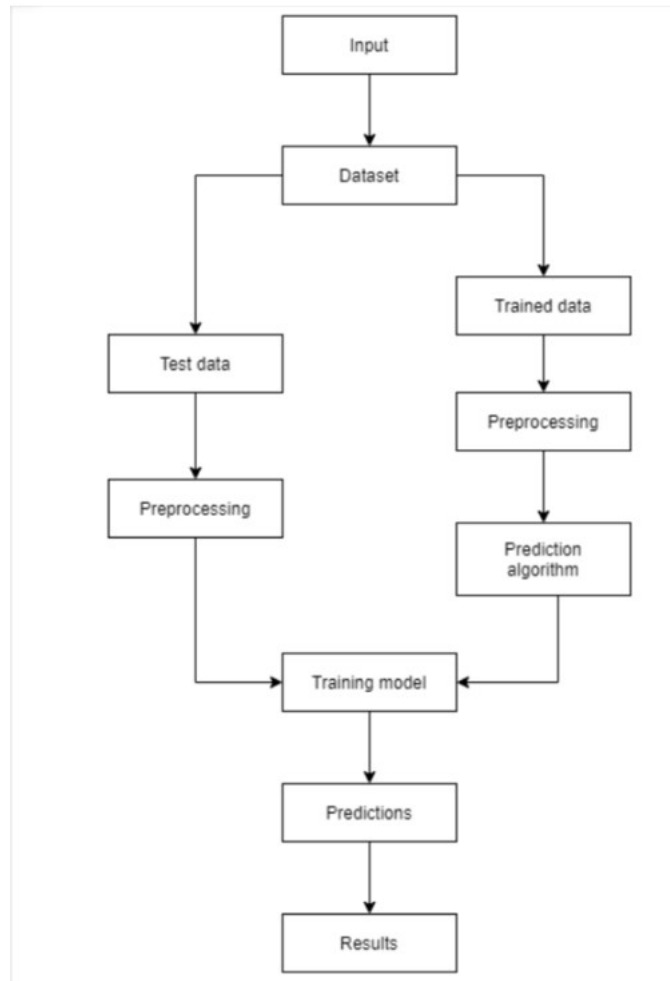


Fig5.1 Architecture Diagram

5.1 Input

For any machine learning project, the input which is the dataset plays a vital role in training of the model with adequate performance. We use a python library called pandas which is used to deal with datasets. In pandas, we use a function called the read () which used to load the dataset.

5.2 Dataset

As previously mentioned, the dataset plays a vital role in model training and the dataset used in this project is Reliance (2020-09 to 2021-09) which is gathered from yahoo finance. The dataset included features such as opening stock price of the day and closing stock price of the day

	Date	Open	High	Low	Close	Adj Close	Volume
0	2020-09-11	2312.000000	2337.850098	2287.100098	2319.750000	2305.370605	27445768
1	2020-09-14	2325.000000	2360.000000	2282.000000	2302.550049	2288.277100	20335273
2	2020-09-15	2311.949951	2325.750000	2288.149902	2318.850098	2304.476318	12542824
3	2020-09-16	2320.000000	2369.350098	2310.550049	2324.550049	2310.140869	15668979
4	2020-09-17	2320.000000	2333.699951	2291.850098	2298.750000	2284.500732	11919927
...	...	...	...	...	...	...	...
242	2021-09-02	2255.000000	2307.800049	2255.000000	2294.399902	2287.512939	4595048
243	2021-09-03	2310.000000	2395.000000	2302.500000	2388.500000	2381.330566	14151629
244	2021-09-06	2413.000000	2480.000000	2412.000000	2425.600098	2418.319092	15525644
245	2021-09-07	2430.000000	2458.000000	2412.000000	2440.899902	2433.572998	8006968
246	2021-09-08	2452.000000	2454.000000	2406.649902	2431.350098	2424.052002	6600210

247 rows × 7 columns

Fig 5.2.1:The Input

5.3 Splitting the data

The dataset is then split into parts for training and testing where the training part is used for training the model and the testing part is used for testing and evaluation the model.

The Dataset is pre-processed before it is actually used this process of pre-processing involves indexing the date so that the time slicing operations can be done much easier.

5.4 Training Model

The training of the model in this project is done with the help of the largest machine learning python library called Sklearn. This library is an open source with wide range of applications in which include Support vector classifier that is used to implement support vector machine technique.

5.5 Output

The final output of this project is a predictor variable that indicates whether to buy a stock or not. And for visual convenience, we used a Python library called Matplotlib, which is used to draw a visual graph of the original returns and predicted returns.

6. IMPLEMENTATION

6.1 Technology Stack

6.1.1 Python

Python is a very popular general-purpose interpreted, interactive, object-oriented, and high-level programming language. Python is dynamically-typed and garbage-collected programming language. It was created by Guido van Rossum during 1985- 1990. Like Perl, Python source code is also available under the GNU General Public License (GPL).

Python supports multiple programming paradigms, including Procedural, Object Oriented and Functional programming language. Python design philosophy emphasizes code readability with the use of significant indentation.

Python is consistently rated as one of the world's most popular programming languages. Python is fairly easy to learn, so if you are starting to learn any programming language then Python could be your great choice. Today various Schools, Colleges and Universities are teaching Python as their primary programming language. There are many other good reasons which makes Python as the top choice of any programmer:

- Python is Open Source which means its available free of cost.
- Python is simple and so easy to learn
- Python is versatile and can be used to create many different things.
- Python has powerful development libraries include AI, ML etc.
- Python is much in demand and ensures high salary

Key advantages of learning Python:

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- Python is Object-Oriented – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- Python is a Beginner's Language – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Applications of Python

The latest release of Python is 3. x. As mentioned previously, Python is one of the most widely used languages on the web. I'm going to list a few of the reasons why that's the case here:

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.

- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.

6.1.2 Jupyter Notebook

Jupyter Notebook is an open-source, web-based interactive environment, which allows you to create and share documents that contain live code, mathematical equations, graphics, maps, plots, visualizations, and narrative text. It integrates with many programming languages like Python, PHP, R, C#, etc.

Advantages of Jupyter Notebook

1. **All in one place:** As you know, Jupyter Notebook is an open-source web-based interactive environment that combines code, text, images, videos, mathematical equations, plots, maps, graphical user interface and widgets to a single document.
2. **Easy to convert:** Jupyter Notebook allows users to convert the notebooks into other formats such as HTML and PDF. It also uses online tools and nbviewer which allows you to render a publicly available notebook in the browser directly.
3. **Easy to share:** Jupyter Notebooks are saved in the structured text files (JSON format), which makes them easily shareable.
4. **Language independent:** Jupyter Notebook is platform-independent because it is represented as JSON (JavaScript Object Notation) format, which is a language-independent, text-based file format. Another reason is that the notebook can be processed by any programming language, and can be converted to any file formats such as Markdown, HTML, PDF, and others.
5. **Interactive code:** Jupyter notebook uses **ipywidgets** packages, which provide many common user interfaces for exploring code and data interactivity.

Disadvantages of Jupyter Notebook

1. It is very hard to test long asynchronous tasks.
2. Less Security
3. It runs cell out of order
4. In Jupyter notebook, there is no IDE integration, no linting, and no code-style correction.

Components of Jupyter Notebook

There are the following three components of Jupyter Notebook -

1. **The notebook web application:** It is an interactive web application for writing and running the code.

The notebook web application allows users to:

- Edit code in the browser with automatic syntax highlighting and indentation.
- Run code on the browser.
- See results of computations with media representations, such as HTML, LaTeX, png, pdf, etc.
- Create and use JavaScript widgets.
- Includes mathematical equations using Markdown cells.

2. **Kernels:** Kernels are the separate processes started by the notebook web application that is used to run a user's code in the given language and return output to the notebook web application.

In Jupyter notebook kernel is available in the following languages:

- Python
- Julia
- Ruby
- R
- Scala
- node.js
- Go

3. **Notebook documents:** Notebook document contains a representation of all content which is visible in the notebook web application, including inputs and outputs of the computations, text, mathematical equations, graphs, and images.

6.1.3 NumPy

NumPy stands for numeric python which is a python package for the computation and processing of the multidimensional and single dimensional array elements.

Travis Oliphant created NumPy package in 2005 by injecting the features of the ancestor module Numeric into another module Numarray.

It is an extension module of Python which is mostly written in C. It provides various functions which are capable of performing the numeric computations with a high speed.

NumPy provides various powerful data structures, implementing multi-dimensional arrays and matrices. These data structures are used for the optimal computations regarding arrays and matrices.

NumPy provides a convenient and efficient way to handle the vast amount of data. NumPy is also very convenient with Matrix multiplication and data reshaping. NumPy is fast which makes it reasonable to work with a large set of data.

The following are advantages of using NumPy for data analysis.

1. NumPy performs array-oriented computing.
2. It efficiently implements the multidimensional arrays.
3. It performs scientific computations.
4. It is capable of performing Fourier Transform and reshaping the data stored in multidimensional arrays.
5. NumPy provides the in-built functions for linear algebra and random number generation.

6.1.4 Pandas

Python Pandas is defined as an open-source library that provides high-performance data manipulation in Python. It is used for data analysis in Python and developed by Wes McKinney in 2008.

Pandas is defined as an open-source library that provides high-performance data manipulation in Python. The name of Pandas is derived from the word Panel Data, which means an Econometrics from Multidimensional data. It is used for data analysis in Python and developed by Wes McKinney in 2008.

Data analysis requires lots of processing, such as restructuring, cleaning or merging, etc. There are different tools available for fast data processing, such as NumPy, SciPy, Cython, and Panda. But we prefer Pandas because working with Pandas is fast, simple and more expressive than other tools.

Pandas is built on top of the NumPy package, means NumPy is required for operating the Pandas.

Before Pandas, Python was capable for data preparation, but it only provided limited support for data analysis. So, Pandas came into the picture and enhanced the capabilities of data analysis. It can perform five significant steps required for processing and analysis of data irrespective of the origin of the data, i.e., load, manipulate, prepare, model, and analyse.

Key Features of Pandas

- It has a fast and efficient Data Frame object with the default and customized indexing.
- Used for reshaping and pivoting of the data sets.
- Group by data for aggregations and transformations.
- It is used for data alignment and integration of the missing data.
- Provide the functionality of Time Series.
- Process a variety of data sets in different formats like matrix data, tabular heterogeneous, time series.
- Handle multiple operations of the data sets such as sub setting, slicing, filtering, group By, re-ordering, and re-shaping.

- It integrates with the other libraries such as SciPy, and scikit-learn.
- Provides fast performance, and If you want to speed it, even more, you can use the **Cython**.

Benefits of Pandas

The benefits of pandas over using other language are as follows:

- **Data Representation:** It represents the data in a form that is suited for data analysis through its Data Frame and Series.
- **Clear code:** The clear API of the Pandas allows you to focus on the core part of the code. So, it provides clear and concise code for the user.

6.1.5 Matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

- Create publication quality plots.
- Make interactive figures that can zoom, pan, update.
- Customize visual style and layout.
- Export to many file formats.
- Embed in JupyterLab and Graphical User Interfaces.
- Use a rich array of third-party packages built on Matplotlib.

Matplotlib is an amazing visualization library in Python for 2D plots of arrays. Matplotlib is a multi-platform data visualization library built on NumPy arrays and designed to work with the broader SciPy stack. It was introduced by John Hunter in the year 2002. One of the greatest benefits of visualization is that it allows us visual access to huge amounts of data in easily digestible visuals. Matplotlib consists of several plots like line, bar, scatter, histogram etc.

6.1.6 Sklearn

A Python module called Scikit-learn offers a variety of supervised and unsupervised learning techniques. It is based on several technologies you may already be acquainted with, including NumPy, pandas, and Matplotlib.

French research scientist David Cournapeau's scikit-learn is a Google Summer of Code venture where the scikit-learn project first began. Its name refers to the idea that it's a modification to SciPy called "SciKit" (SciPy Toolkit), which was independently created and published. Later, other programmers rewrote the core codebase.

The French Institute for Research in Computer Science and Automation at Rocquencourt, France, led the work in 2010 under the direction of Alexandre Gramfort, Gael Varoquaux, Vincent Michel, and Fabian Pedregosa. On February 1st of that year, the institution issued the project's first official release. In November 2012, scikit-learn and scikit-image were cited as examples of scikits that were "well-maintained and popular". One of the most widely used machine learning packages on GitHub is Python's scikit-learn.

Scikit-learn is mainly coded in Python and heavily utilizes the NumPy library for highly efficient array and linear algebra computations. Some fundamental algorithms are also built in Cython to enhance the efficiency of this library. Support vector machines, logistic regression, and linear SVMs are performed using wrappers coded in Cython for LIBSVM and LIBLINEAR, respectively. Expanding these routines with Python might not be viable in such circumstances.

Scikit-learn works nicely with numerous other Python packages, including SciPy, Pandas data frames, NumPy for array vectorization, Matplotlib, seaborn and plotly for plotting graphs, and many more.

Key concepts and features include:

- Algorithms for making decisions, such as:
 - Data are identified and categorised by classification as per the patterns.
 - Regression is the process of forecasting or predicting data values using the historical and anticipated data average.
- Clustering is the automatic collection of datasets with related data.
 - Predictive analysis is supported by various algorithms, including neural networks for pattern recognition and straightforward linear regression.
 - Compatibility with the libraries of NumPy, pandas, and matplotlib
- A predictive model can be built or trained on input data by computers using machine learning (ML), eliminating the need for explicit programming. A subset of AI is machine learning (AI).

6.2 Sample code:

```
# Machine learning
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

# For data manipulation
import pandas as pd
import numpy as np

# To plot
import matplotlib.pyplot as plt
plt.style.use('seaborn-darkgrid')

# To ignore warnings
```

```

import warnings
warnings.filterwarnings("ignore")
df = pd.read_csv('RELIANCE.csv')
df

# Changes The Date column as index columns
df.index = pd.to_datetime(df['Date'])
df

# drop The original date column
df = df.drop(['Date'], axis='columns')
df

# Create predictor variables
df['Open-Close'] = df.Open - df.Close
df['High-Low'] = df.High - df.Low

# Store all predictor variables in a variable X
X = df[['Open-Close', 'High-Low']]
X.head()

# Target variables
y = np.where(df['Close'].shift(-1) > df['Close'], 1, 0)
y
split_percentage = 0.8
split = int(split_percentage*len(df))

# Train data set
X_train = X[:split]
y_train = y[:split]

# Test data set

```

```

X_test = X[split:]
y_test = y[split:]

# Support vector classifier
cls = SVC().fit(X_train, y_train)
df['Predicted_Signal'] = cls.predict(X)

# Calculate daily returns
df['Return'] = df.Close.pct_change()

# Calculate strategy returns
df['Strategy_Return'] = df.Return
*df.Predicted_Signal.shift(1)
df['Cum_Ret'] = df['Return'].cumsum()
df

# Plot Strategy Cumulative returns
df['Cum_Strategy'] = df['Strategy_Return'].cumsum()
df
import matplotlib.pyplot as plt
%matplotlib inline
plt.plot(Df['Cum_Ret'],color='red')
plt.plot(Df['Cum_Strategy'],color='blue')

```


7. RESULTS

- The raw results of a machine learning project are the model's accuracy. And for this project, we found an average accuracy of 56%. And a 50+% can be viewed as effective because the stock can only move up or down, which translates to 50% accuracy, and our model, which gives an accuracy of 56%, is much better.
- The Fig 7.1 shows the dataset which is date indexed and with predictive variable (Predicted_Signal) which is essentially what model predicts for the new data

	Open	High	Low	Close	Adj Close	Volume	Open-Close	High-Low	Predicted_Signal
Date									
2020-09-11	2312.000000	2337.850098	2287.100098	2319.750000	2305.370605	27445768	-7.750000	50.750000	1
2020-09-14	2325.000000	2360.000000	2282.000000	2302.550049	2288.277100	20335273	22.449951	78.000000	0
2020-09-15	2311.949951	2325.750000	2288.149902	2318.850098	2304.476318	12542824	-6.900147	37.600098	1
2020-09-16	2320.000000	2369.350098	2310.550049	2324.550049	2310.140869	15668979	-4.550049	58.800049	1
2020-09-17	2320.000000	2333.699951	2291.850098	2298.750000	2284.500732	11919927	21.250000	41.849853	1
...	...	...	...	...	...	...	...	...	...
2021-09-02	2255.000000	2307.800049	2255.000000	2294.399902	2287.512939	4595048	-39.399902	52.800049	1
2021-09-03	2310.000000	2395.000000	2302.500000	2388.500000	2381.330566	14151629	-78.500000	92.500000	0
2021-09-06	2413.000000	2480.000000	2412.000000	2425.600098	2418.319092	15525644	-12.600098	68.000000	1
2021-09-07	2430.000000	2458.000000	2412.000000	2440.899902	2433.572998	8006968	-10.899902	46.000000	1

Fig 7.1 Predictive variable

- The Fig 7.2 shows the accuracy of the model trained with SVM technique in this Project:

```
#Import svm model
from sklearn import svm

#Create a svm Classifier
clf = svm.SVC(kernel='linear') # Linear Kernel

#Train the model using the training sets
clf.fit(X_train, y_train)

#Predict the response for test dataset
y_pred = clf.predict(X_test)

#Import scikit-learn metrics module for accuracy calculation
from sklearn import metrics

# Model Accuracy: how often is the classifier correct?
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

Accuracy: 0.56

Fig 7.2 Accuracy

- The Graph below show the visual representation of the accuracy (Original Result vs. Predicted Result).

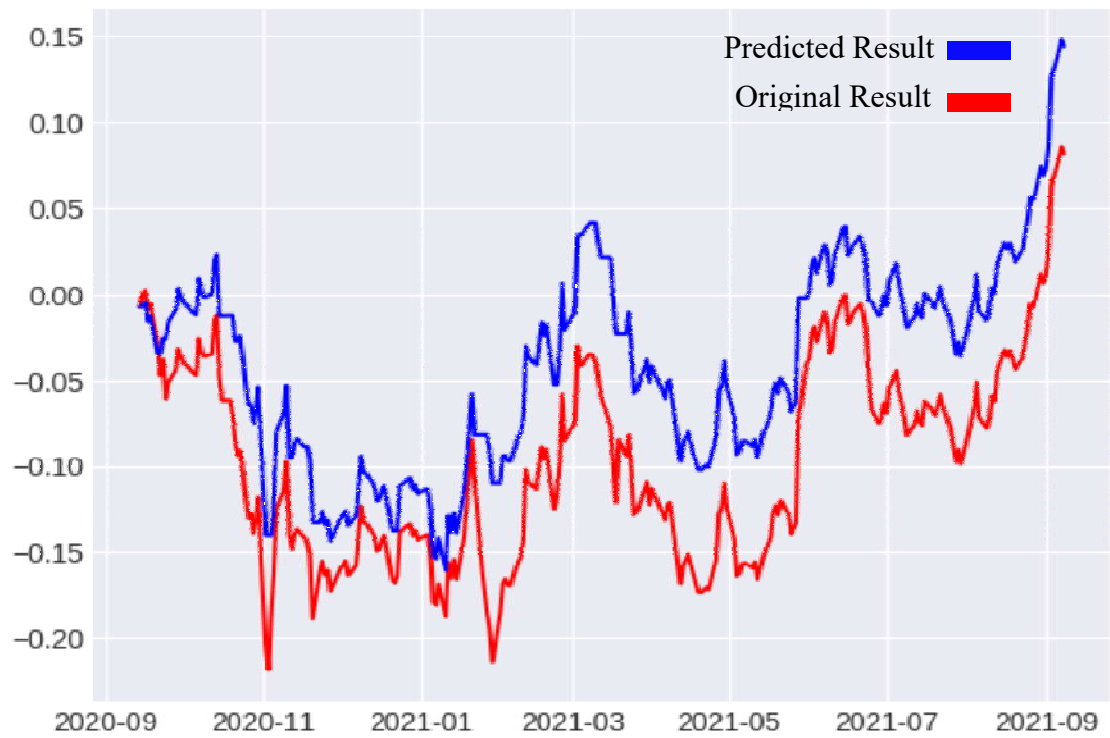


Fig 7.3 Output Result

8. CONCLUSION

This project used historical data of Reliance industries limited spanning from 09-2020 to 09-2021 the dataset in question consist of features such as date, close time stock price, opening time stock price, peak stock price of the day and the lowest stock price. The train-split method was used in this project. This project, which was built with the objective of finding out the predictive performance of the SVM model, has proven to be effective, as can be seen by its 56% average accuracy. And for a machine learning model to be considered effective, it must have an accuracy of 50+%, and this model can be improved in the future by using a dataset that gives prices of stocks on a minute or second basis as opposed to the daily stock prices used in this project.

Future Scope:

The result of this project turned out to be fruitful but using the splitting phase and model it is possible for a user to find out the abnormal events which drastically affected the stock market. The workings of this process involves finding the accuracy of the model using the same dataset while incrementing the split percentage by 5% each time. If the accuracy is increasing while the split percentage is increasing then it means that the stock market did not experience any kind of abnormal event which caused bearish moment in the stock market but if there is a decrease in accuracy then it means that the stock market did experience an event due to which a bearish movement can be seen.

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